

Naveen Khan

AI Engineer · Machine Learning · Generative AI · Full-Stack AI

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B.E. Computer Systems Engineering · Mehran UET (11/2021 – 12/2025)

SUMMARY

AI Engineer with 6 months of professional experience plus a 12-month research project. I design and build AI systems across machine learning, generative AI (RAG, LLMs), computer vision and AI automation — turning complex data, documents and workflows into useful, well-engineered products. Strong full-stack background in Python, FastAPI, Next.js, PostgreSQL and vector databases (Faiss, Pinecone).

EXPERIENCE

Full Stack AI Engineer · AI Hackathon · Sofstica Solutions (Pvt.) Ltd · Karachi · 08 / 2026 · 48-Hour Solo Hackathon Project

Built a clinical research platform with Text-to-SQL capabilities across 912,284 patient records — entirely solo within a 48-hour hackathon sprint.

- Solo-built clinical research platform within a 48-hour hackathon sprint.
- Engineered end-to-end data-quality engine from scratch with full provenance tracking.
- Integrated FastAPI + Next.js app with LLM-powered natural language interfaces.
- Enabled real-time cohort exploration, analysis, and AI-generated summaries.
- Tools: Python, FastAPI, Next.js, SQLite, LLM, LangChain, Docker, Azure.

AI Engineer Intern · ITSolera Pvt. Ltd. · Karachi · 01 / 2026 – 04 / 2026

Trained CNN and YOLO-based computer vision models on 10,000+ images across 3 domains.

- Fine-tuned models to 93%+ accuracy through systematic hyperparameter optimization.
- Domains: medical image classification, damaged road detection, theft detection.
- Automated repetitive data workflows, cutting manual processing time by ~40%.
- Tools: Python, Deep Learning, FastAPI, Streamlit, Roboflow, Computer Vision.

AI Engineer Intern · Civil Aviation Authority of Pakistan (CAA) · Karachi · 07 / 2025 – 08 / 2025

Engineered enterprise-level conversational system using LLMs and RAG.

- Reduced internal query resolution time by 60% via LLM-powered conversational system.
- Architected semantic retrieval framework with RAG across 100+ organizational PDFs.
- Implemented embedding-based document indexing for fast retrieval.
- Tools: Python, FastAPI, LangChain, RAG, NLP, LLMs, Hugging Face, .NET.

AI Research · Multimodal Smart Wearable for Personal Safety · Mehran UET · 11 / 2024 – 11 / 2025

Developed AI-powered wearable with YOLO-based robbery detection, multilingual speech recognition, and GPS/GSM emergency response.

- Achieved 95% accuracy in real-time threat detection with alert response under 5 seconds.
- Implemented multilingual speech recognition + evidence capturing on-device.
- Engineered GPS/GSM emergency response pipeline on Raspberry Pi 4 + ESP32.
- Won 2nd Place at IEEE CS Exhibition 2025 (45+ projects); published paper in 2026.
- Tools: Python, PyTorch, YOLOv8, Raspberry Pi 4, ESP32, IoT, GPS/GSM, Computer Vision.

SELECTED PROJECTS

01 · ClinData Explorer · Healthcare AI

AI-powered clinical cohort and data-quality explorer · Text-to-SQL · 912K records · Live on Azure.

02 · McDonald's AI Agent · Customer AI

AI customer-support and order-management workflow · LLM · FastAPI.

03 · CardioRisk AI · Clinical Decision Support

Heart-disease risk prediction · 97.6% accuracy (SVM) · 3,800+ patients · Next.js + Python + Vercel.

04 · Enterprise RAG Assistant · Document Intelligence

Retrieval-augmented generation system for enterprise document intelligence · Vector DB · LLM.

05 · Radiomed · Clinical AI

Automated medical image diagnosis assistant · CNN + Densenet · 99.8% top accuracy · Brain MRI / Chest X-ray / Lung CT · Flask + SQL + HTML/CSS/JS.

06 · SAFELINK · *Multimodal Wearable*

Multimodal smart wearable for personal safety · Computer Vision + Edge AI + IoT.

07 · Revenue AI · *Sales Forecasting*

Sales forecasting platform · Polynomial Regression · 95.3% R^2 · 0.903 MAE · Streamlit · Live on Vercel.

08 · Lead Generation Agent · *AI Automation*

AI-powered lead generation agent · n8n + OpenRouter LLM · 50+ personalized emails/day · \$0.02/email.

SKILLS

Generative AI: RAG · LLMs · LangChain · Prompt Engineering · AI Agents · Vector Databases (Pinecone, FAISS)

Machine Learning: Classification · Regression · Feature Engineering · EDA · Model Evaluation

Computer Vision: YOLO · Image Classification · Object Detection · Roboflow · Data Augmentation

Engineering: Python · FastAPI · Streamlit · MySQL / PostgreSQL · HTML / CSS / JS · Git / GitHub

AI Automation: n8n · AI Agents · Webhooks · LLM API Integration · Workflow Automation · JSON Handling

Infrastructure: Docker · Vercel · Azure · GCP

EDUCATION

BE Computer Systems Engineering · *Mehran University of Engineering & Technology, Pakistan* · 11 / 2021 – 12 / 2025

PUBLICATIONS

• Khan, N., et al. (2026). *SAFELINK: A Multimodal Smart Wearable Device for Personal Safety*. THESES: International Journal of Multidisciplinary Research.

ACHIEVEMENTS

- Built clinical research platform with Text-to-SQL across 912,284 patient records at Sofstica AI Hackathon.
- Trained CNN + YOLO models on 10,000+ images with 93%+ accuracy at ITSolera.
- Reduced enterprise query resolution time by 60% via LLM + RAG system at CAA.
- Won 2nd Place at IEEE CS Exhibition 2025 (among 45+ projects) for SAFELINK research.
- Published SAFELINK research paper globally in 2026.
- BE Computer Systems Engineering, Mehran UET (11/2021 – 12/2025).